EMPLOYEE LEAVE MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

BHARATHWAJ S

(240701078)

in partial fulfilment of the End Semester Examination for the course

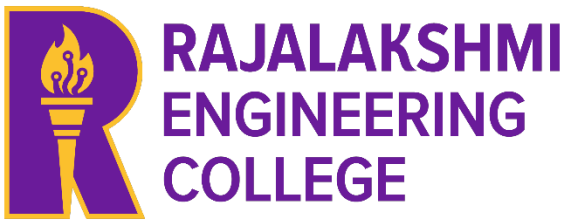
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING

COMPUTER SCIENCE AND ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

NOVEMBER 2026

RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled "**EMPLOYEE LEAVE MANAGEMENT SYSTEM**" is the bonafide work of **BHARATHWAJ S (240701078)**, who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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EXTERNAL EXAMINER

DECLARATION

I, BHARATHWAJ S (240701078), hereby declare that the project report titled “EMPLOYEE LEAVE MANAGEMENT SYSTEM”, submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of the project work carried out by me individually. The application described in this report was developed using the Zoho Creator low-code platform and is based on the application screens and project information documented in this report.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Any external resource, platform documentation, sample material or AI-assisted content used in preparing this report should be acknowledged where applicable.

Place: Chennai
Date: _____

SIGNATURE OF THE STUDENT
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I also acknowledge Zoho Corporation for providing access to the Zoho Creator platform and learning resources, and I thank my family and friends for their unwavering support.

BHARATHWAJ S

ABSTRACT

Employee leave management is an important administrative activity in an organisation. Employees need a structured way to submit leave requests, while the organisation must maintain employee records, review requests, track approval decisions and monitor leave balances. When these activities are handled through paper forms, email messages or separate spreadsheets, information can become scattered, request status may be difficult to track, and leave records may require repeated manual updates.

This project presents an Employee Leave Management System developed using the Zoho Creator low-code platform. The application provides a dashboard and dedicated sections for leave applications, employee registrations, departments, approvals, leave types, balances, policies, documents, summaries and Zoho People employee records. The leave application form captures employee information, employee ID, department, leave type, start date, end date, number of days and reason for leave. The application also includes approval records with employee name, leave category, date range, number of days and approval status.

The dashboard displays leave-related summary cards, a chart comparing rejected and approved counts, AI processing status information and quick links for creating leave applications, employee registrations and leave balance records. The screenshots supplied for this report show the application interface and sample records. These records are treated as demonstration data; their displayed totals and employee names are not presented as verified organisational statistics.

The project demonstrates how a low-code platform can be used to organise leave-related information through forms, reports, navigation and dashboard components. The current report documents the functionality visible in the supplied application screenshots. Features such as automatic balance deduction, notifications, approval workflows, integrations and security configuration are discussed as possible design considerations only where their actual implementation has not been confirmed.

Keywords: Employee Leave Management; Zoho Creator; Low-Code Development; Leave Application; Employee Records; Approval Tracking; Dashboard; Business Process Automation

PROJECT AT A GLANCE

Attribute	Details
Project Title	Employee Leave Management System
Application Domain	Human Resources / Employee Administration
Student Name & Register No.	BHARATHWAJ S – 240701078
Course Faculty	Ms. Jinusophia
Platform	Zoho Creator (as shown in the application screenshots)
Application Sections	Dashboard; Applications; Employee Registrations; Approvals; Balances; Policies; Documents; Summaries; Zoho People Employees
Leave Application Fields	Employee; Employee ID; Department; Leave Type; Start Date; End Date; Number of Days; Reason for Leave
Screenshots Included	Dashboard, Leave Application form, Employee Registration report, Approvals report and Zoho People Employees report
Deployment Status	Not specified in the supplied screenshots
Implementation Notes	Workflow logic, notification rules and live integration details require confirmation from the actual application configuration.

TABLE OF CONTENTS

ABSTRACT..... 5
PROJECT AT A GLANCE.....6
LIST OF TABLES.....9
LIST OF FIGURES.....10
LIST OF ABBREVIATIONS.....11
INTRODUCTION.....12
1.1 Background.....12
1.2 Problem Statement.....12
1.3 Objectives.....12
1.4 Scope of the Project.....13
1.5 Justification for a Low-Code Approach.....13
1.6 Organisation of the Report.....14
LITERATURE SURVEY AND SYSTEM STUDY.....15
2.1 Study of the Existing System.....15
2.2 Limitations of the Existing System.....15
2.3 Low-Code versus Traditional Development.....15
2.4 Literature Survey.....16
2.5 Emerging Trends in Low-Code Development.....17
2.6 Proposed System.....17
REQUIREMENT ANALYSIS.....18
3.1 Stakeholders and User Roles.....18
3.2 Functional Requirements.....18
3.3 Non-Functional Requirements.....19
3.4 Hardware and Software Requirements.....19
3.5 Use Case Model.....20
SYSTEM DESIGN.....21
4.1 System Architecture.....21
4.2 Data Model.....21
4.3 Form Design.....22
4.4 Process and Workflow Design.....23
4.5 User Interface and Navigation Design.....23
4.6 Access Control Design.....23
IMPLEMENTATION.....24
5.1 Application Setup.....24
5.2 Forms and Data Capture.....24

5.3 Reports....26
5.4 Pages and Dashboards....26
5.5 Deluge Scripting....27
5.6 Workflows and Automation....27
5.7 Integrations....28
5.8 User Management and Role-Based Access....28
5.9 AI and Analytics Features....28
5.10 Pro-Code Extension....28
5.11 Mobile and Portal Access....29
TESTING AND DEBUGGING....30
6.1 Testing Strategy....30
6.2 Test Cases....31
6.3 Debugging and Validation....31
6.4 Defect Log....31
6.5 User Acceptance Testing....31
DEPLOYMENT, SECURITY AND GOVERNANCE....32
7.1 Environments and Deployment Strategy....32
7.2 Performance Optimisation....32
7.3 Security Features....32
7.4 Governance and Administration....32
RESULTS AND DISCUSSION....33
8.1 Application Output....33
8.2 Achievement of Objectives....33
8.3 Comparison with the Existing System....33
8.4 Limitations of the Low-Code Approach....34
8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum....34
CONCLUSION AND FUTURE ENHANCEMENTS....35
9.1 Conclusion....35
9.2 Future Enhancements....35
9.3 Learning Outcomes and Career Relevance....36
REFERENCES....37
APPENDIX A – APPLICATION FEATURE CHECKLIST....38
APPENDIX B – COURSE OUTCOME MAPPING....39
APPENDIX C – IMPLEMENTATION NOTES....40
APPENDIX D – USER MANUAL....41
APPENDIX E – EVALUATION RUBRIC....43
APPENDIX F – VIVA-VOCE RECORD SHEET....45

LIST OF TABLES

Table 2.1: Low-Code and Traditional Development Comparison.....16
Table 2.2: Literature Survey Summary.....16
Table 3.1: Stakeholders and User Roles.....18
Table 3.2: Functional Requirements.....18
Table 3.3: Non-Functional Requirements.....19
Table 3.4: Hardware and Software Requirements.....19
Table 3.5: Leave Application Use Case
Table 4.1: Application Modules and Data.....22
Table 4.2: Leave Application Field Specification.....22
Table 4.3: Employee Registration Field Specification
Table 4.4: Approval Record Field Specification
Table 4.5: Dashboard Components
Table 6.1: Test Cases.....30
Table 6.2: Defect Log.....31
Table 8.1: Objective Achievement.....33
Table E.1: Evaluation Rubric for the Individual Project.....43
Table E.2: Performance Level Descriptors.....44
Table F.1: Viva-Voce Questions and Responses.....45
Table F.2: Marks Summary.....45

LIST OF FIGURES

Figure 3.1: Use Case Model.....20
Figure 4.1: System Architecture.....21
Figure 4.2: Conceptual Data Model
Figure 5.1: Employee Leave Management System Dashboard.....25
Figure 5.2: Leave Application Form.....25
Figure 5.3: Employee Registration Report.....25
Figure 5.4: All Approvals Report.....26
Figure 5.5: Zoho People Employees Report.....26

LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
HR	Human Resources
ID	Identifier
LCAP	Low-Code Application Platform
NFR	Non-Functional Requirement
RBAC	Role-Based Access Control
SDLC	Software Development Life Cycle
UI	User Interface
UAT	User Acceptance Testing

CHAPTER 1
INTRODUCTION

1.1 Background

Leave management is a routine but important function in human resource administration. Employees may request casual leave, sick leave, earned leave or other leave categories depending on organisational policy. The request must be recorded with the employee identity, department, date range, reason and required approval status.

In a manual process, leave requests may be submitted through paper forms, email or messaging. HR staff or managers then check employee details, calculate the leave duration, review leave balances and communicate the decision. When the information is stored in separate places, it becomes difficult to maintain a single, reliable view of leave requests and employee records.

The Employee Leave Management System is designed to centralise these records in a Zoho Creator application. The supplied screenshots show a navigation structure that separates leave applications, employee registration, approvals, balances, policies, documents and summaries. This structure makes the application easier to navigate and gives the administrator a consolidated view of leave-related activities.

1.2 Problem Statement

Organisations need an orderly method to receive and review employee leave requests. A fragmented or manual process can create delays, inconsistent records, difficulty checking leave balances and limited visibility into the status of requests.

The project addresses the need for a central application where employee details and leave requests can be recorded, viewed and tracked. The leave request form captures the information needed to identify the employee, classify the leave, determine the requested dates and record the reason. The approvals section provides a structured report of request details and decision status.

1.3 Objectives

The primary objective is to develop a low-code application that organises employee leave records and supports a clear leave-request workflow. The specific objectives are:

- To provide a form for entering employee leave applications with employee, department, leave type, date range, number of days and reason.
- To maintain employee registration records and department information.
- To provide an approvals report containing leave request details and approval status.
- To provide navigation to leave types, balances, policies, documents and summaries.
- To display leave-related information through dashboard cards, charts and quick links.
- To demonstrate the use of Zoho Creator forms, reports and pages in a practical administrative application.

1.4 Scope of the Project

The scope documented in this report is limited to the features visible in the supplied screenshots: dashboard metrics and charts; leave application data entry; employee registration records; approval records; and a Zoho People Employees section. The application menu also contains leave types, departments, balances, policies, documents and summaries.

The screenshots do not show all form fields, workflow configuration screens, validation scripts, notification setup or integration settings. Therefore, those implementation details are not claimed as verified. They may be added to the report if the user provides further screenshots or project configuration details.

Out of scope for this report are payroll processing, attendance tracking, biometric integration, statutory compliance automation and external HR-system synchronisation beyond the Zoho People-related section visible in the application.

1.5 Justification for a Low-Code Approach

Zoho Creator provides visual form building, reports, pages and application navigation in a single environment. This reduces the amount of infrastructure and interface code required for a small administrative application. A developer can configure fields, create reports and arrange pages using a visual builder, then extend the application with Deluge scripts when additional business rules are required.

Low-code development is suitable for this project because the core operations involve structured records, form submission, report viewing and dashboard presentation. The platform allows the application to be adjusted as leave categories, departments or organisational procedures change. However, low-code does not remove the need for careful data modelling, testing, access control and validation.

1.6 Organisation of the Report

Chapter 1 introduces the project and its objectives. Chapter 2 studies the existing leave-management process and the proposed low-code approach. Chapter 3 presents requirements and user roles. Chapter 4 describes the conceptual design. Chapter 5 documents the application interface and visible implementation. Chapter 6 discusses testing. Chapter 7 covers deployment, security and governance considerations. Chapter 8 evaluates the application based on the supplied screenshots. Chapter 9 concludes the work and identifies future enhancements. Appendices provide a feature checklist, user guidance and evaluation material.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In a conventional leave-management process, an employee submits a request through a paper form, email or an online form. The request is reviewed by a supervisor or HR representative. The reviewer checks the employee's details, leave category, requested dates and available balance before approving or rejecting the request. The decision is then communicated to the employee and recorded for future reference.

Employee details and leave history may be maintained in spreadsheets or separate HR records. When a manager needs to know how many requests are pending, how many were rejected or how much leave an employee has used, the information may need to be collected from multiple sources. This increases administrative effort and creates the risk of inconsistent records.

2.2 Limitations of the Existing System

- Leave requests may be stored in different formats and locations.
- Manual entry can introduce spelling errors, duplicate records and incorrect dates.
- Employees may not have a convenient way to view request status.
- Supervisors may have limited visibility into pending or rejected applications.
- Leave balances can become inconsistent if they are updated manually.
- Generating summary reports may require additional spreadsheet work.
- Historical leave information may be difficult to search and audit.

2.3 Low-Code versus Traditional Development

Traditional development typically requires separate work for database design, user-interface development, server-side logic, hosting and deployment. A low-code platform offers built-in components for forms, reports and application navigation. The developer focuses on configuring the business data and processes, while custom scripting can be used for rules that cannot be handled through configuration alone.

Table 2.1: Low-Code and Traditional Development Comparison

Aspect	Traditional Development	Low-Code Approach
Forms and UI	Designed and coded separately	Configured using visual form builder
Data storage	Database schema and backend integration required	Platform-managed form data and field types
Reports	Custom queries and UI components	Built-in report views and filters
Business logic	Backend code and services	Visual workflows and Deluge scripting
Deployment	Application hosting and release pipeline	Platform-managed application environment
Flexibility	High control, but more code to maintain	Fast configuration; platform constraints apply

2.4 Literature Survey

Low-code development has become a practical approach for building internal business applications, workflow tools and data-driven forms. Research and industry material commonly emphasise visual modelling, reusable platform components, shorter development cycles and the importance of governance. At the same time, platform dependence, customisation limits and the need for secure configuration remain important considerations.

Zoho Creator documentation is the primary technical reference for platform forms, reports, pages, workflows and Deluge scripting. Since the report is based on screenshots rather than a full exported application specification, the implementation claims in later chapters are restricted to the visible features.

Table 2.2: Literature Survey Summary

Source	Focus	Relevance
Zoho Creator product documentation	Forms, reports, pages, workflow configuration and platform capabilities	Primary reference for understanding the platform components used
Low-code development literature	Visual development, productivity and platform abstraction	Supports the choice of a low-code platform for structured administrative workflows
Human-resource information system concepts	Employee records, leave requests and approval tracking	Provides the business context for the application modules

2.5 Emerging Trends in Low-Code Development

Current low-code platforms increasingly combine visual development with automation, analytics, AI-assisted configuration and integrations. Dashboard components can summarise operational data, while workflow tools can reduce repetitive administrative tasks. Organisations also need to consider access control, data governance, auditability and the ability to maintain applications over time.

The supplied application dashboard includes numerical summary cards, a chart and AI processing status. The screenshot establishes that these visual elements are present, but it does not establish the specific AI model, underlying processing rules or whether the displayed values represent production activity.

2.6 Proposed System

The proposed system provides a central Zoho Creator application for employee leave administration. It organises the process into separate modules: applications for leave requests, employee registrations for employee records, approvals for review outcomes, leave types for categories, balances for leave entitlement information, policies for rules, documents for supporting records, summaries for consolidated views and Zoho People Employees for employee-related information.

The dashboard serves as the landing page and provides quick links to create common records. The application aims to reduce scattered records and improve the visibility of leave-related information. Any automatic leave balance calculation, approval notification or synchronisation should be considered a further verified feature only after inspecting its workflow configuration.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

The application is intended to support the people involved in employee leave administration. The exact access permissions configured in the live application were not shown in the screenshots; the following roles describe the logical users of the system.

Table 3.1: Stakeholders and User Roles

Role	Description	Expected responsibilities
Employee	Person requesting leave	Submit leave request details and provide the reason for leave
Approver / Manager	Person responsible for reviewing a request	Review request dates and leave category; record approval or rejection
HR / Administrator	Maintains employee and leave records	Maintain employee details, leave types, balances, policies and reports
Application Owner	Zoho Creator application administrator	Configure forms, reports, navigation, access and workflows

3.2 Functional Requirements

The functional requirements below are based on the modules and fields visible in the supplied screenshots. Requirements for automation are labelled as expected or proposed where the screenshots do not prove their implementation.

Table 3.2: Functional Requirements

ID	Requirement	Priority
FR-01	Provide a leave application form with employee, employee ID, department, leave type, start date, end date, number of days and reason.	High
FR-02	Provide an employee registration module and an all-employee registration report.	High
FR-03	Provide an approvals module with leave request, employee name, leave type, start date, end date, number of days and approval status.	High
FR-04	Provide navigation to leave types and departments.	High
FR-05	Provide a section for leave balances.	High
FR-06	Provide policies, documents and summaries sections.	Medium
FR-07	Provide a dashboard with leave-related cards, chart visualisation and quick links.	High
FR-08	Provide access to a Zoho People Employees section as shown in the application navigation.	Medium
FR-09	Support recording and reviewing leave approval decisions.	High

ID	Requirement	Priority
FR-10	Provide date and leave-duration validation; exact rule implementation must be confirmed in the application.	High

3.3 Non-Functional Requirements

Non-functional requirements describe the quality expected from the application, including usability, accuracy, data integrity, security and maintainability. The following are design requirements rather than claims that each has already been tested.

Table 3.3: Non-Functional Requirements

ID	Category	Requirement
NFR-01	Usability	Forms should use clear labels and suitable field types; navigation should be understandable.
NFR-02	Accuracy	Date ranges and leave-day values should be consistent and checked before approval.
NFR-03	Data integrity	Employee identity and leave request details should remain linked consistently.
NFR-04	Availability	The application should be accessible through the supported Zoho Creator environment.
NFR-05	Security	Employee information should be visible only to authorised users.
NFR-06	Maintainability	Forms, reports, fields and workflows should use consistent names and documentation.
NFR-07	Auditability	Approval status changes and leave-related updates should be traceable where supported.

3.4 Hardware and Software Requirements

The application is accessed through Zoho Creator. Development and use require a supported web browser, an internet connection and a Zoho account with access to the application. A laptop or desktop is suitable for application configuration and report preparation.

Table 3.4: Hardware and Software Requirements

Category	Requirement
Hardware	Laptop or desktop computer; adequate memory and display resolution for browser-based development
Network	Reliable internet connection
Operating System	Windows, macOS or Linux supported by a modern browser
Browser	Google Chrome or another modern browser supported by Zoho Creator
Platform	Zoho Creator

Category	Requirement
Scripting	Deluge, if custom workflows or validations are configured
Storage	Zoho Creator application storage; local storage for report screenshots and project files

3.5 Use Case Model

The conceptual use-case model includes an employee submitting a leave request, an approver reviewing the request and an HR administrator maintaining employee and leave records. The administrator can review application records and dashboard summaries. Figure 3.1 presents the main interactions at a high level.



Figure 3.1: Conceptual Use Case Model

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

The system follows a platform-based architecture. Users access the application through a web browser. The Zoho Creator interface provides the navigation menu, forms, reports and dashboard page. Form data is stored by the platform, while reports present records in tabular form. Workflow logic can be attached to form events when configured by the application developer.

Figure 4.1 shows the conceptual structure. The exact backend infrastructure and configured workflow events are managed by Zoho Creator and are not directly visible in the screenshots.

Employee / Approver / HR Administrator
Web Browser – Zoho Creator Application
Forms • Reports • Dashboard • Navigation
Zoho Creator Data Storage and Configured Workflows

Figure 4.1: Conceptual System Architecture

4.2 Data Model

The visible application indicates several data groups: employee information, leave applications, approval records, leave types, departments and leave balances. The approvals report includes a leave-request identifier, employee name, leave type, start and end dates, number of days and approval status. The employee registration report includes employee ID, employee name, email address, phone number, department and designation, with additional fields extending beyond the screenshot viewport.

The conceptual data model links leave applications to an employee record. A leave type classifies the request, and an approval record captures the decision. Department information helps organise employees. Leave balances are associated with employees and leave categories. The exact lookup relationships and internal field names should be verified from the Zoho Creator form builder before being documented as implemented.

Table 4.1: Application Modules and Data

Module / Entity	Visible or expected data	Purpose
Employee Registration	Employee ID, name, email, phone, department, designation and additional employee details	Maintain employee master information
Leave Application	Employee, Employee ID, Department, Leave Type, Start Date, End Date, Number of Days, Reason for Leave	Capture a leave request
Approval Record	Leave Request, Employee Name, Leave Type, Start Date, End Date, Number of Days, Approval Status	Review and track leave decisions
Leave Types	Leave category and related configuration (fields not shown)	Define categories such as casual, sick or earned leave
Departments	Department records (fields not shown)	Organise employee information
Balances	Employee leave balance records (fields not shown)	Maintain entitlement and usage information
Policies / Documents / Summaries	Module names visible; detailed fields not shown	Store policy information, related documents and summary views

4.3 Form Design

The leave application form is designed as a structured data-entry screen. It contains the following fields as seen in Screenshot 2. The employee and employee ID fields identify the applicant. Department and leave type classify the request. Start and end dates define the requested period. Number of days represents the duration, and the reason field captures the employee's explanation.

Table 4.2: Leave Application Field Specification

Field	Visible field type / appearance	Purpose
Employee	Text / selection field (appearance in screenshot)	Identify the employee submitting the request
Employee ID	Text field	Record the employee's identifier
Department	Text field	Identify the employee's department
Leave Type	Dropdown	Choose the leave category
Start Date	Date field (dd-MMM-yyyy)	First day of requested leave
End Date	Date field (dd-MMM-yyyy)	Last day of requested leave
Number of Days	Numeric field	Record requested leave duration
Reason for Leave	Multiline text area	Explain the purpose of the leave

4.4 Process and Workflow Design

The basic leave process consists of employee registration, leave application submission, application review and approval-status recording. A request begins with the employee details and selected leave type. The request includes the start date, end date and duration. An approver reviews the information and records an approval or rejection decision.

Possible workflow checks include verifying that the end date is not earlier than the start date, ensuring that required fields are completed, calculating the requested duration and checking leave balance before approval. These are recommended rules; the screenshots do not show the workflow editor or Deluge implementation, so the report does not state that these rules are already automated.

4.5 User Interface and Navigation Design

The left navigation panel is the primary way to move between application modules. It contains Dashboard, Applications, Employee Registrations, Approvals, Balances, Policies, Documents, Summaries and Zoho People Employees. Some modules expand to show subpages, such as Leave Applications, All Applications and Leave Types under Applications; Employee Registrations, All Employee Registrations and Departments under Employee Registrations; and Leave Approvals, All Approvals and Leave Types under Approvals.

The dashboard uses cards and quick links to make common operations easy to reach. The green highlight identifies the active navigation item. The interface uses a consistent side navigation and content area across the screenshots.

4.6 Access Control Design

Employee leave records contain personal and employment information. A production application should distinguish employee, approver and HR administrator permissions. Employees should be able to create and review their own requests; approvers should review requests assigned to them; HR administrators should manage employee master data and reporting. These are recommended access-control principles. Actual user roles, sharing settings and permissions must be checked in Zoho Creator before claiming that they are configured.

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application is titled “Employee Leave Management System” in the Zoho Creator interface. The application was configured with a left-side navigation menu and a dashboard landing page. The screenshots show the application running in the Zoho Creator environment, with application editing and sharing controls available at the top of the screen.

The following subsections describe the visible modules and interface elements. The report does not invent Deluge code, workflow counts, exact platform edition or deployment URLs because those details were not included in the supplied screenshots.

5.2 Forms and Data Capture

The Leave Applications form captures the employee and leave details. Its visible fields are Employee, Employee ID, Department, Leave Type, Start Date, End Date, Number of Days and Reason for Leave. The leave type is selected from a dropdown, date fields use a dd-MMM-yyyy display format, and the reason is entered in a multiline area.

Employee registration is maintained separately. The All Employee Registrations report shows five sample records in the screenshot. The visible columns include Employee ID, Employee Name, Email Address, Phone Number, Department and Designation. Some additional columns are partially outside the screenshot, so they are not transcribed as fully visible fields.

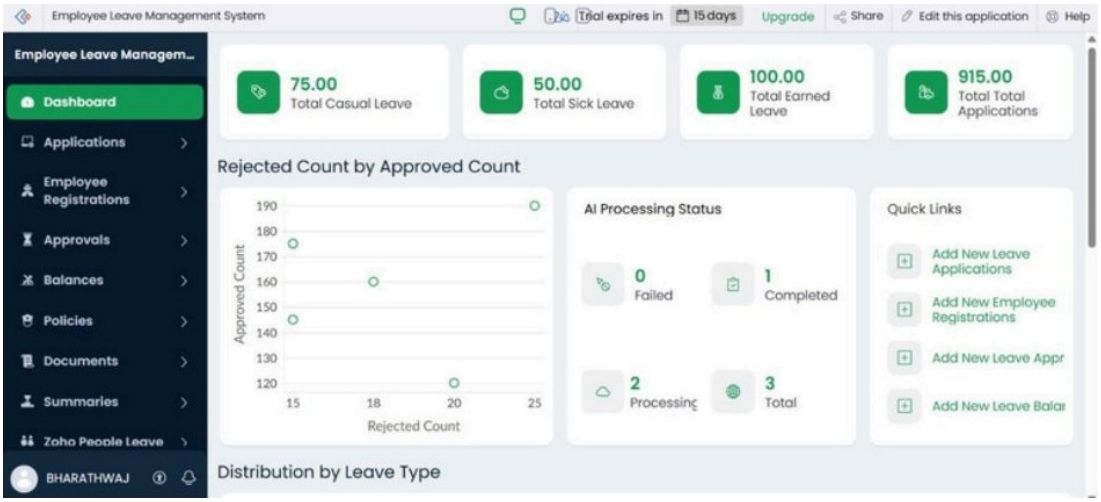


Figure 5.1: Employee Leave Management System – Dashboard

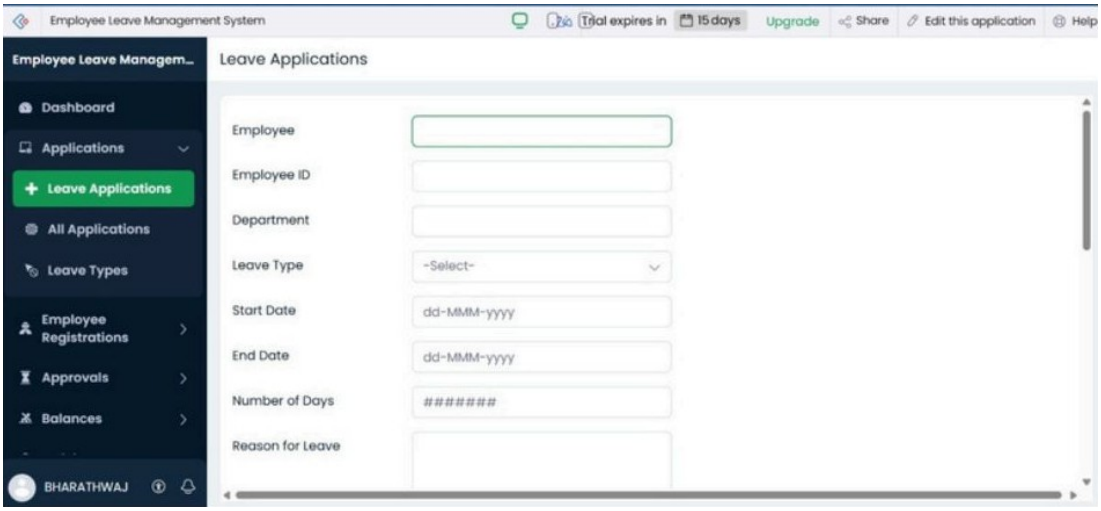


Figure 5.2: Leave Applications – Data Entry Form

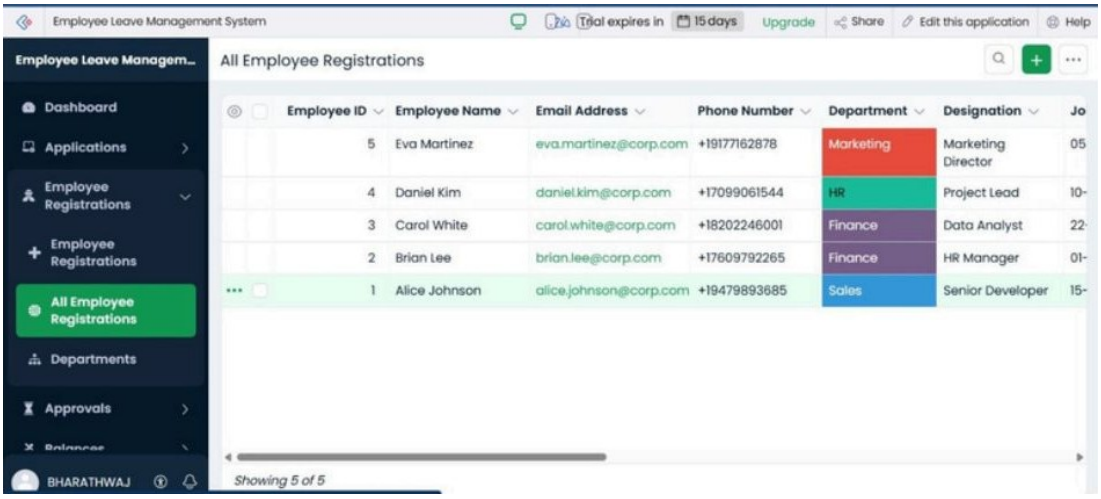


Figure 5.3: All Employee Registrations – Report

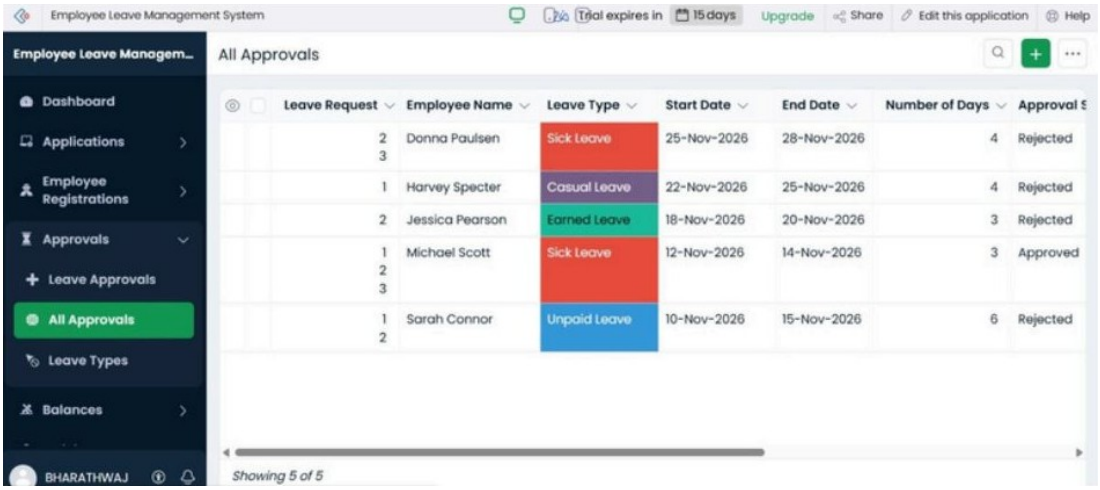


Figure 5.4: All Approvals – Report

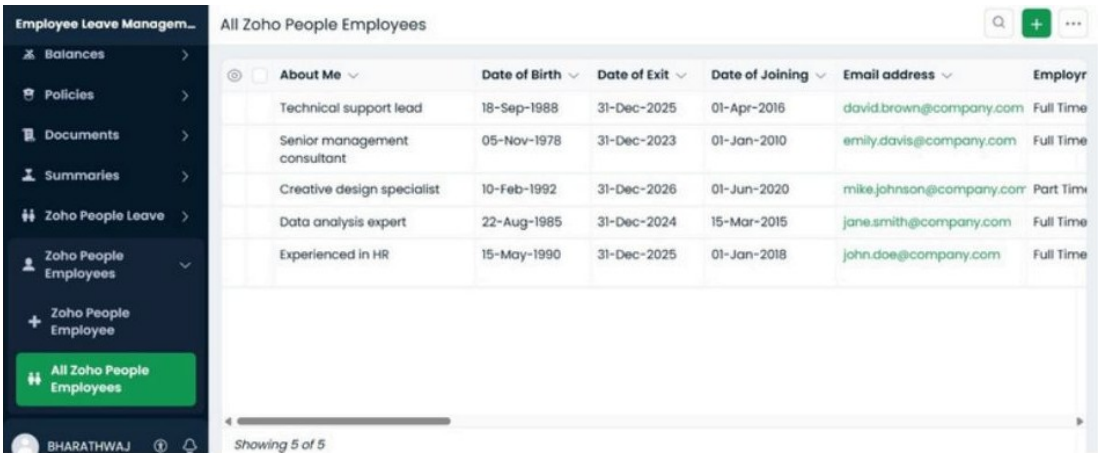


Figure 5.5: All Zoho People Employees – Report

5.3 Reports

The application contains tabular reports for viewing records. The screenshot of All Employee Registrations displays five records and shows the employee ID, employee name, email address, phone number, department and designation. The All Approvals screenshot displays five sample leave records with leave type, dates, number of days and approval status.

Reports provide a convenient way to inspect records without opening each form individually. In a complete implementation, reports can be configured with filters, sorting and role-based visibility. Specific filter settings are not visible in the screenshots and are therefore not described as implemented.

5.4 Pages and Dashboards

The Dashboard is the application's overview page. The top row shows four summary cards: Total Casual Leave (75.00), Total Sick Leave (50.00), Total Earned Leave

(100.00) and Total Total Applications (915.00). The displayed label “Total Total Applications” is reproduced as it appears in the screenshot; it may be a label typo in the application UI.

The dashboard includes a chart titled “Rejected Count by Approved Count,” an AI Processing Status panel and a Quick Links panel. The chart displays plotted points and axes labelled Rejected Count and Approved Count. The AI Processing Status panel shows Failed, Completed, Processing and Total counters. Quick links include Add New Leave Applications, Add New Employee Registrations, Add New Leave Applications and Add New Leave Balances (the last two labels are partially clipped in the screenshot).

These displayed metrics are screenshot values from the application's sample or current data. They are not interpreted as verified business statistics. The dashboard is useful as a central entry point and a visual summary of application data.

5.5 Deluge Scripting

Zoho Creator supports Deluge scripting for custom validation, calculations and workflow actions. In an employee leave application, Deluge may be used to calculate leave duration, validate date ranges, check required fields, update balances after approval or send notifications. These examples describe potential uses of Deluge and not confirmed code in the current application. No complete Deluge script was supplied in the screenshots. Therefore, this report does not include invented source code or claim that specific scripts have been executed. The actual script source should be exported from the application and added to this section if required by the course evaluation.

5.6 Workflows and Automation

Leave-management workflows can automate repetitive steps such as routing an application for approval, updating the status after a decision, notifying the employee and updating leave balance records. A workflow should be designed carefully so that rejected requests do not reduce the employee's balance and duplicate updates do not occur.

The screenshots show application and approval modules but do not show workflow configuration or notification logs. Workflow implementation status, number of workflows and trigger events are therefore not stated as verified facts.

5.7 Integrations

The application navigation contains a section named Zoho People Employees, and the supplied screenshot shows an employee list with fields such as About Me, Date of Birth, Date of Exit, Date of Joining and Email Address. This establishes that a Zoho People Employees section is present in the interface. The screenshot alone does not prove whether records are synchronised live through an integration, imported manually or generated as sample data.

Any integration with Zoho People or another HR platform should be documented with the actual connection settings, data mapping and sync configuration. Those details were not provided.

5.8 User Management and Role-Based Access

Employee leave data should be protected because it may include personal information and reasons for absence. Role-based access is recommended for separating employee, approver and HR administrator responsibilities. Access should be tested using accounts with different permissions. The current screenshots do not show the application's permission profiles or user-role configuration.

5.9 AI and Analytics Features

The Dashboard contains a panel labelled AI Processing Status with Failed, Completed, Processing and Total counters. The presence of this panel is visible in the screenshot. However, the screenshots do not identify the AI service, the processing task, the data source or the logic used to generate these counts. The report therefore describes the panel as a visible UI feature without making claims about its underlying AI capability.

The dashboard chart and summary cards represent analytics-style visualisations. In a deployed application, the data source and aggregation logic should be checked to ensure the figures are meaningful and up to date.

5.10 Pro-Code Extension

Zoho Creator can be extended through Deluge and, where supported, APIs or external services. Potential extensions for this project include advanced approval routing, calendar integration, audit logging and synchronisation with HR systems. These are future extensions rather than features confirmed by the screenshots.

5.11 Mobile and Portal Access

Zoho Creator applications may be accessible through supported mobile experiences, depending on platform configuration and plan. The supplied screenshots are desktop browser captures. Mobile layout, employee self-service portal configuration and portal permissions were not shown and are not claimed as implemented.

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing should confirm that forms accept valid data, reject invalid input, preserve relationships between employee and leave records and display approval outcomes correctly. The current screenshots provide evidence that the dashboard, leave application form, employee report, approvals report and Zoho People Employees report render in the application. They do not provide evidence of a complete test run or test results.

Testing for a leave-management system should include positive cases, negative cases, boundary dates, missing required fields, approval status changes and leave balance consistency. The test cases below are proposed validation scenarios and should be executed against the live application before they are reported as passed.

Table 6.1: Suggested Test Cases

TC ID	Test Scenario	Expected Result	Status
TC-01	Open dashboard	Dashboard and navigation load correctly	Observed in screenshot
TC-02	Open leave application form	All visible form fields are displayed	Observed in screenshot
TC-03	Submit leave request with required fields	Record is saved if required validation passes	Not verified
TC-04	Leave end date earlier than start date	Application should be prevented or flagged	Not verified
TC-05	Leave type not selected	Required-field validation should be displayed	Not verified
TC-06	Enter valid date range	Number of days should match configured calculation rules	Not verified
TC-07	Review approval list	Leave request and approval status are displayed	Observed in screenshot
TC-08	Approve a request	Status should update and be visible in report	Not verified
TC-09	Reject a request	Status should update and be visible in report	Not verified
TC-10	Check employee registration report	Employee records and visible columns render	Observed in screenshot
TC-11	Open Zoho People Employees section	Employee list renders	Observed in screenshot
TC-12	Check balance after approved leave	Balance should update according to policy	Not verified

6.2 Test Cases

Test cases are designed around the main user actions. Dashboard and report display checks can be supported by the supplied screenshots. Form submission, validation, approval updates and leave-balance calculation require direct execution in the application and should be recorded after testing.

6.3 Debugging and Validation

Common issues in a leave application include incorrect date formats, missing employee identifiers, inconsistent leave duration, incorrect lookup values and duplicate approval records. Debugging should begin by reproducing the issue, checking the form field configuration, reviewing workflow execution and confirming that the record values are correct.

Where Deluge is used, logs and workflow execution details should be inspected. Validation messages should be clear and should tell the user what needs to be corrected.

6.4 Defect Log

Table 6.2: Defect Log Template

Defect ID	Issue	Severity	Resolution	Status
D-01	Date-range validation to be checked	Medium	Verify start/end date rule in form or workflow	Open / not verified
D-02	Number-of-days calculation to be checked	High	Compare calculated duration with leave policy	Open / not verified
D-03	Dashboard card label appears as “Total Total Applications”	Low	Review dashboard display label and correct if intended	Observed UI label
D-04	AI Processing Status data source is undocumented	Medium	Confirm what process the panel represents	Open / not verified

6.5 User Acceptance Testing

User acceptance testing should be performed by an employee, approver and HR administrator. The participants should confirm that the form labels are understandable, requests can be reviewed, status values are clear and leave records are easy to find. Acceptance should be recorded only after actual users complete the test scenarios and approve the results.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application is shown in the Zoho Creator environment with application editing and sharing controls. The screenshots do not identify a published production URL, development/stage environment configuration or deployment date. A production release should include a review of form fields, reports, workflow rules, permissions and backup/export arrangements before users are given access.

7.2 Performance Optimisation

Performance can be improved by keeping forms focused, avoiding unnecessary lookup fields, limiting dashboard queries to relevant records and using reports with appropriate filters. Dashboard counters should be based on clearly defined criteria. These recommendations are relevant to the application design but were not performance-tested in the supplied screenshots.

7.3 Security Features

Employee leave information should be protected against unauthorised access. Security measures should include account authentication, role-based permissions, limited access to sensitive employee details, secure sharing and careful handling of exported records. Leave reasons may contain personal information, so only authorised staff should view them.

The screenshots show the Zoho Creator application interface but do not expose permission settings, audit logs or security test results. This section therefore describes recommended security practices rather than claiming that every control has been configured.

7.4 Governance and Administration

Application governance includes consistent naming, clear ownership, change tracking, regular review of leave categories and policies, and testing before workflow changes are released. The organisation should define who can add leave types, modify employee data, approve requests and view summary reports. Changes to leave rules should be reflected in both the application configuration and the documented policy.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The screenshots demonstrate the following visible outputs: a dashboard with leave summary cards and analytics-style panels; a leave application form with eight visible fields; an employee registration report showing five records; an approvals report showing five records and approval statuses; and a Zoho People Employees report showing five records.

The dashboard's displayed cards show Casual Leave 75.00, Sick Leave 50.00, Earned Leave 100.00 and Total Applications 915.00 (the UI label is shown as “Total Total Applications”). The approval report includes sample requests with Casual Leave, Sick Leave, Earned Leave and Unpaid Leave, with dates in November 2026 and statuses including Rejected and Approved. These are the values visible in the supplied screenshots and should be treated as sample data rather than real institutional records.

8.2 Achievement of Objectives

Table 8.1: Objective Achievement

Objective	Evidence from supplied screenshots	Assessment
Provide leave application entry	Leave Applications form displays employee, ID, department, type, date range, number of days and reason	Interface evidence available
Maintain employee registration	All Employee Registrations report shows employee records and fields	Interface evidence available
Track approval decisions	All Approvals report displays leave type, dates, days and status	Interface evidence available
Provide central dashboard	Dashboard shows cards, chart, AI status panel and quick links	Interface evidence available
Organise related HR modules	Navigation includes leave types, departments, balances, policies, documents, summaries and Zoho People Employees	Interface evidence available
Automate balance updates and notifications	No workflow configuration or execution screenshots supplied	Not verified

8.3 Comparison with the Existing System

The application provides a more structured interface than paper forms or separate spreadsheets. Leave requests and employee information are organised into dedicated modules, and approval records can be viewed in a report. The dashboard gives a quick visual summary, while the navigation links users to common operations.

However, the degree of improvement in processing time, error reduction or employee satisfaction cannot be quantified without a baseline and a measured evaluation. No such quantitative study was supplied for this report.

8.4 Limitations of the Low-Code Approach

Low-code platforms simplify the development of forms and reports, but the application remains dependent on platform features, plan limits and configuration. Complex approval policies may require custom scripts or integrations. Data security, permission design and testing remain the responsibility of the application developer.

Current report limitations include the absence of workflow screenshots, complete field configuration, exact integration settings, access-control details and a verified test log. The application screenshots also contain demonstration-style employee names and numerical values. These limitations are documented rather than hidden.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

The visible application components—forms, reports, navigation and dashboard cards—are primarily visual configuration tasks. Deluge scripting would represent the low-code extension layer for validations, calculations and automation. External APIs or custom services would represent a more pro-code extension. The project illustrates how a structured administrative application can begin with visual components and expand when requirements demand additional logic.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Employee Leave Management System is a Zoho Creator application intended to organise employee leave requests and related HR records. The supplied screenshots show a dashboard, leave application form, employee registration report, approval report and Zoho People Employees section. The application navigation also includes leave types, departments, balances, policies, documents and summaries.

The project demonstrates the value of a low-code approach for developing a structured administrative application. Forms provide a consistent way to capture leave details, reports support record review and the dashboard provides a central overview. The report has documented the visible implementation while distinguishing screenshot evidence from features that require additional verification.

9.2 Future Enhancements

- Configure a complete employee self-service leave request process.
- Add approval routing based on department, reporting manager or leave category.
- Automatically calculate leave duration according to the organisation's calendar and holiday policy.
- Deduct leave balance only after approval and restore it when a request is cancelled or rejected, where policy requires.
- Send email or in-app notifications for submission, approval, rejection and cancellation.
- Add calendar views and team leave planning.
- Configure role-based permissions for employees, approvers and HR administrators.
- Add audit logs for leave status and balance changes.
- Verify and document the Zoho People integration and synchronisation process.

- Add report filters for department, employee, leave category, status and date range.
- Test the application with real user scenarios and maintain a formal test-results log.

9.3 Learning Outcomes and Career Relevance

The project provides practical experience in modelling employee records, designing data-entry forms, arranging reports and dashboards, and understanding the workflow of a human-resource application. It also introduces the importance of validation, data integrity, access control and testing in low-code development.

These skills are relevant to business application development, HR information systems, workflow automation and low-code application development roles. Further work on Deluge, integration configuration and security testing would deepen the technical implementation.

REFERENCES

[1] Zoho Corporation, “Zoho Creator – Low-Code Application Development Platform,” official product and help resources. https://www.zoho.com/creator/

[2] Zoho Corporation, “Zoho Creator Help – Forms, Reports and Pages,” official documentation. https://help.zoho.com/portal/en/kb/creator

[3] Zoho Corporation, “Deluge – Zoho’s Scripting Language,” official documentation. https://www.zoho.com/deluge/

[4] Zoho Corporation, “Zoho People,” official product information. https://www.zoho.com/people/

[5] Project screenshots supplied by the student: Employee Leave Management System dashboard, leave application form, employee registration report, approvals report and Zoho People Employees report (November 2026).

[6] Laundry Service Management System Project Report, supplied sample report, used as a formatting and chapter-structure reference.

APPENDIX A – APPLICATION FEATURE CHECKLIST

A.1: Feature Checklist

Feature / Module	Visible in supplied screenshots?	Notes
Dashboard	Yes	Summary cards, chart, AI Processing Status and Quick Links
Leave Applications	Yes	Form fields visible
All Applications	Menu item visible	Report content not separately captured
Leave Types	Menu item visible	Detailed fields not shown
Employee Registrations	Yes	Registration form/report navigation visible
Departments	Menu item visible	Detailed fields not shown
Approvals	Yes	Leave approval report visible
Balances	Menu item visible	Detailed fields not shown
Policies	Menu item visible	Detailed fields not shown
Documents	Menu item visible	Detailed fields not shown
Summaries	Menu item visible	Detailed fields not shown
Zoho People Employees	Yes	Employee report visible; integration mechanism not confirmed

APPENDIX B – COURSE OUTCOME MAPPING

B.1: Indicative Course Outcome Mapping

Project Activity	Relevant learning	Evidence
Form design	Translate requirements into structured fields	Leave Application screenshot
Report configuration	Present stored data in a readable tabular view	Employee Registration and Approvals screenshots
Dashboard design	Summarise data using cards and charts	Dashboard screenshot
Application navigation	Organise modules for usability	Left navigation screenshots
System analysis and documentation	Identify scope, requirements, limitations and improvements	Chapters 1–9
Testing and quality	Define validation and test scenarios	Chapter 6 test plan

APPENDIX C – IMPLEMENTATION NOTES

The following implementation details should be added after reviewing the Zoho Creator application builder. They are intentionally left as verification items rather than filled with invented values.

C.1: Items to Verify in the Application

Item	Required evidence
Form names and exact field link names	Form builder screenshots or exported form specification
Leave duration calculation	Field properties or Deluge script
Approval workflow	Workflow editor screenshot and trigger configuration
Leave balance update	Workflow source and test record
Notifications	Email action configuration and received notification
User permissions	Role/profile settings and permission matrix
Zoho People connection	Integration configuration, mapping and sync status
Application publication	Published URL and deployment settings
Test execution	Completed test cases with actual pass/fail results

APPENDIX D – USER MANUAL

D.1 Accessing the Application

1. Open a modern web browser and sign in to the Zoho Creator account that has access to the application.
2. Open the Employee Leave Management System application.
3. Use the left navigation panel to select Dashboard, Applications, Employee Registrations, Approvals or another available module.
4. Use the dashboard quick links to access common record-entry forms.

D.2 Submitting a Leave Application

1. Open Applications → Leave Applications.
2. Enter or select the employee information.
3. Enter the Employee ID and Department.
4. Select the required Leave Type.
5. Choose the Start Date and End Date.
6. Enter the Number of Days according to the application's configured rules.
7. Enter the Reason for Leave.
8. Review the information and submit the form. The exact submission button and confirmation message should be confirmed in the live application.

D.3 Reviewing Employee Registrations

1. Open Employee Registrations.
2. Select All Employee Registrations to view the employee report.
3. Review the employee ID, name, email address, phone number, department and designation columns.
4. Use the report's available search or filters if configured.

D.4 Reviewing Leave Approvals

1. Open Approvals → All Approvals.
2. Locate the leave request using the employee name, leave type or dates.

3. Review the requested duration and approval status.
4. If the application provides an approval action, use the configured action to record the decision. The precise approval procedure should be confirmed from the application's workflow.

D.5 Viewing Dashboard and HR Information

Open Dashboard to view summary cards, chart, AI Processing Status and Quick Links. Use Balances, Policies, Documents and Summaries to access their respective modules. Open Zoho People Employees to view the employee-related list shown in the screenshot.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	BHARATHWAJ S	**Register No.**	240701078
Project Title	Employee Leave Management System	**Date**	_____

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Problem clarity, objectives, functional and non-functional requirements	10	
2	System Design	Architecture, data model, form and workflow design	10	
3	Forms, Reports & Pages	Field types, reports, navigation and dashboard design	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts and automation (evaluate only if implemented)	15	
5	Testing & Debugging	Test coverage, validation and evidence of results	15	
6	Security & Governance	Permissions, data protection and application administration	10	
7	Documentation	Report structure, screenshots, clarity and completeness	15	
8	Viva-Voce	Explanation of design decisions and project limitations	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner

Name: _____

Signature of External Examiner

Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	BHARATHWAJ S	**Register No.**	240701078
Department / Section	Computer Science and Engineering	**Date of Viva**	_____
Application URL Verified	Yes / No	**Demo Duration**	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–7 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

INTERNAL EXAMINER

Name: _____

EXTERNAL EXAMINER

Name: _____